

ព្រះរាជាណាចក្រកម្ពុជា
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INSTITUTE OF TECHNOLOGY OF CAMBODIA

Internet Programming I

Project Proposal

Group: I4-GIC-B

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1. Purpose and scope

ShopStock is a web application for a small shop. Staff record every product that comes in or goes out, and the manager sees current stock, low-stock alerts and monthly reports.

1.1 Problem. Small shops in Phnom Penh often track stock in a notebook or spreadsheet. This causes wrong counts because changes are not written down (P1), stock-outs that nobody sees coming (P2), and no record of who changed what (P3). ShopStock solves P1 with a movement record for every change, P2 with low-stock alerts, and P3 with a per-user movement history that cannot be edited.

1.2 Scope. In scope: two roles with login, products with categories and suppliers, stock in, stock out and adjustments, low-stock alerts, a monthly report, search, filter and pagination, and automated tests. Out of scope: point-of-sale screen, payments, purchase orders, barcode scanning, several branches and notifications.

2. User roles

ShopStock has two roles with different permissions.

Role	Who	What they can do
Staff	Shop assistants and storekeepers	Search products, record stock in and stock out, see their own movement history. A self-registered account starts inactive until a manager activates it.
Manager	The shop owner or manager	Everything staff can do, plus manage products, categories and suppliers, adjust stock after a count, see all movements, low-stock alerts and the monthly report, and activate staff accounts.

3. Functional requirements

The functional requirements say what ShopStock shall do. The last column links each one to the user story in section 9 that it serves.

3.1 Features

ID	The system shall…	Story
AUTH-1	let a visitor register with name, email and password, creating an inactive Staff account	US-01
AUTH-2	reject a duplicate email (409) or a password shorter than 8 characters (400)	US-01
AUTH-3	store passwords only as bcrypt hashes and never return them in a response	US-01
AUTH-4	log in an active user and return a JWT valid for 1 hour; an inactive user gets 403	US-02
AUTH-5	answer a wrong email or a wrong password with the same message	US-02
AUTH-6	redirect a visitor without a token to /login and hide actions the role cannot use	US-02
AUTH-7	let a manager activate or deactivate staff accounts	US-10

ID	The system shall…	Story
CAT-1	let a manager create, view, update and delete categories and suppliers	US-10
CAT-2	refuse to delete a category or supplier that still has products	US-10
PRD-1	let a manager create a product (SKU, name, category, supplier, unit, cost price, sale price, reorder level) starting at quantity 0	US-07
PRD-2	require a unique SKU in the format AAA-0000 and a sale price not lower than the cost price	US-07
PRD-3	show any logged-in user the product list and a detail page with quantity, stock value and the last 10 movements	US-03
PRD-4	let a manager update a product's fields, except quantity, which changes only through movements	US-07
PRD-5	let a manager delete a product without movements, or archive one with movements	US-07
PRD-6	search products by name or SKU, filter by category, supplier and low stock, and page results (10 per page, 50 maximum)	US-03
MOV-1	let staff and managers record stock in (PURCHASE, CUSTOMER_RETURN)	US-04
MOV-2	let staff and managers record stock out (SALE, DAMAGED, EXPIRED, RETURN_TO_SUPPLIER)	US-05
MOV-3	let a manager record an adjustment from a counted quantity, with a required note	US-08
MOV-4	save each movement and update the quantity in one database transaction, never below 0	US-04, US-05
MOV-5	not allow any movement to be edited or deleted	US-06
MOV-6	show staff only their own movements and a manager all movements, with filters	US-06
REP-1	show the manager a low-stock list and a count badge in the menu	US-09
REP-2	show the manager a monthly report: stock in and out per category, top 5 products sold, total stock value	US-09
UI-1	show API 400, 404 and 409 errors as readable messages next to the form or at the top of the page	all

3.2 Business rules (enforced in the backend service, not only in the user interface)

ID	Rule	Response
BR-1	A stock in or out quantity is a whole number from 1 to 10,000	400 · Quantity must be a whole number from 1 to 10,000
BR-2	A stock out cannot exceed the current quantity	409 · Not enough stock: only N left
BR-3	An archived product receives no movements	409 · This product is archived
BR-4	The reason must match the movement type	400 · Reason X is not allowed for type Y
BR-5	Adjustments are manager-only and need a note and a real difference	403 / 400 · No difference to adjust
BR-6	SKU is unique and matches AAA-0000	409 / 400 · SKU already exists
BR-7	Sale price is not lower than cost price	400 · Sale price cannot be lower than cost price
BR-8	A product with movements can only be archived, not deleted	409 · Archive it instead
BR-9	A category or supplier with products cannot be deleted	409 · Move or delete its products first

3.3 User interface: eleven screen groups, each with a wireframe in docs/wireframes/ at Check 1.

Route	Screen	Role
/login, /register	Log in, register	public
/products	Product list with search, filters, pagination and low-stock badges	both
/products/:id	Product detail with stock in / out form and last 10 movements	both
/my-movements, /movements/:id	My movements and movement detail (read-only)	staff (own)
/manager/products/new, /:id/edit	Product form (create, edit, archive)	manager
/manager/adjust/:productId	Stock count adjustment	manager
/manager/alerts	Low-stock alerts	manager
/manager/dashboard	Monthly report and stock value	manager
/manager/movements	All movements with filters	manager
/manager/staff, /categories, /suppliers	Staff accounts, categories, suppliers	manager
404 and /forbidden	Not found and forbidden pages	all

4. Non-functional requirements

Each non-functional requirement has a measurable target and a way to check it.

ID	Requirement	How it is checked
SEC-1	Passwords hashed with bcrypt (cost 10); JWT signed with JWT_SECRET from .env, which is never committed	Database check; .gitignore
SEC-2	Global ValidationPipe with whitelist: unknown or invalid fields return 400	API test
SEC-3	Wrong role returns 403; another user's movement returns 404 (IDOR protection)	API tests
SEC-4	CORS allows only the frontend origin; no response contains passwordHash	Manual check; API test
PERF-1	The product list with 2,000 products returns one page in under 500 ms	Timed request on seeded data
REL-1	Data survives a backend restart; quantity never goes below 0, even with two requests at once	Restart test; parallel API test
REL-2	Money stored as Decimal(10,2), never as a JavaScript float	Schema review
USE-1	Recording a stock out takes at most 3 actions from the product page	Manual walk-through
USE-2	Layouts work from 360 px to 1,440 px wide; every list has a loading and an empty state	Browser dev tools; component tests
MAIN-1	TypeScript strict mode; npm run build passes with no type errors; one NestJS module per resource	CI
PORT-1	A new developer runs both apps from a fresh clone in under 10 minutes with the README	A classmate tries it

5. System architecture

ShopStock has three tiers: a Vue single-page app in the browser, a NestJS REST API, and a PostgreSQL database reached only through Prisma.

5.1 Components

Component	Responsibility
Vue views, Pinia stores, router guard	Screens in 3.3; one store per entity; the guard sends visitors without a token to /login
src/api/http.ts	Axios with VITE_API_URL; adds the Authorization header; turns API errors into messages
Guards, ValidationPipe, controllers	Return 401 or 403, reject invalid bodies with 400, map routes to services
Services	Apply the business rules, ownership checks and transactions
Prisma and PostgreSQL 16	The only database access; stores all data (section 6)

5.2 Interfaces. The API uses HTTP with JSON and has 23 endpoints in seven modules: auth, users, categories, suppliers, products, movements and reports. Every endpoint except register and login needs a Bearer token. Errors use the shape { statusCode, message, error }. The full contract with bodies and statuses goes in docs/API.md at Check 3.

5.3 Software and hardware requirements

Part	Requirement
Frontend	Vue 3 + TypeScript with Vite, Pinia, Vue Router, Axios; tests with Vitest
Backend	NestJS 11 on Node.js 22 LTS; class-validator, @nestjs/jwt, passport-jwt, bcrypt; tests with Jest + Supertest
Database	PostgreSQL 16 (local or Docker) with Prisma ORM and committed migrations; separate test database in backend/.env.test
Development machine	Windows, macOS or Linux laptop with at least 8 GB RAM, Node.js 22 and Git
User device	Phone, tablet or computer with a current Chrome, Firefox or Edge browser

6. Data management

6.1 Entities. Five entities in PostgreSQL through Prisma, with four one-to-many relations: Category → Products, Supplier → Products, Product → Movements, User → Movements.

Entity	Main fields	Rules
User	fullName, email, passwordHash, role (STAFF, MANAGER), isActive	Email unique; self-registered staff start inactive
Category	name, description	Name unique
Supplier	name, phone, email, address	Name unique
Product (main)	sku, name, categoryId, supplierId, unit, costPrice, salePrice, quantity, reorderLevel, isActive	Full CRUD; quantity ≥ 0, changed only by movements
StockMovement	productId, userId, type (IN, OUT, ADJUST), reason, change, quantityAfter, note, createdAt	Never edited or deleted

6.2 Storage, backup and recovery. The product keeps a quantity column, and every change is also a movement row, so the history explains the number. Every schema change is a committed Prisma migration, and npx prisma db seed creates demo data with one user per role. During development, data is re-created with prisma migrate reset and the seed; when deployed, a daily pg_dump is kept for 7 days and restored with pg_restore. Tests use a separate database in backend/.env.test.

7. User documentation

README.md explains the purpose, setup from a fresh clone, environment variables, seeded accounts, test commands and project map. docs/USER_GUIDE.md gives step-by-step tasks for staff and for the manager. In the app, fields show format hints ("SKU, e.g. DRK-0001"), errors use the exact messages in 3.2 and section 9, and empty lists say what to do next.

8. Testing requirements

API end-to-end tests use Jest + Supertest (npm run test:e2e) on a separate test database that is reset and seeded before each file. Component tests use Vitest (npm run test:unit). GitHub Actions runs both on every push from Check 6, and docs/TEST_PLAN.md gives one test name per acceptance criterion. The key flows tested:

Flow	Expected result	Story
Log in as an inactive staff user	403, waiting for approval	US-02
Search "coca", page 1	Only matches, at most 10 rows	US-03
Stock in 10, then stock out 4	Quantity +10 then −4, two movements	US-04, US-05
Stock out more than in stock	409, quantity unchanged	US-05
Two parallel stock outs of the last unit	One succeeds, one gets 409	US-05
Staff B opens staff A's movement	404 (IDOR case)	US-06
Staff creates a product or an adjustment	403	US-07, US-08
Request without a token	401	all
Stock-out form with quantity 0 (component)	Message shown, no API call	US-05
Product list with a low-stock product (component)	Badge on that row only	US-03

9. Acceptance criteria: user stories

A story is Done only when all its acceptance criteria pass in the tagged commit. There are 10 stories across the 2 roles.

9.1 Story overview

ID	Story	Role	Planned check
US-01	Register	Staff	Check 5
US-02	Log in	Both	Check 5
US-03	Find a product (search, filter, pagination)	Staff	Checks 2, 4, 6
US-04	Record stock in	Staff	Checks 2, 4
US-05	Record stock out	Staff	Checks 2, 4
US-06	See my movements	Staff	Checks 4, 5
US-07	Manage products	Manager	Checks 3, 4
US-08	Adjust stock after a count	Manager	Check 5
US-09	Low-stock alerts and monthly report	Manager	Check 7
US-10	Manage staff, categories and suppliers	Manager	Checks 4, 5

9.2 User stories

US-01 Register (Staff)
As a new shop assistant, I want to create an account so that the manager can give me access.

AC	Acceptance criterion
AC1	With a valid name, email and a password of 8 or more characters, the account is created as inactive Staff and I see "Account created. Wait for the manager to activate it".
AC2	An email already in use shows "This email is already registered".
AC3	An invalid email or a password shorter than 8 characters shows a message under the field, and nothing is sent to the server.

Status: Not done

US-02 Log in (Both)
As a user, I want to log in so that I see the pages for my role.

AC	Acceptance criterion
AC1	Correct credentials take staff to /products and a manager to /manager/dashboard.
AC2	A wrong email or a wrong password shows the same message, "Invalid email or password", and no token is stored.
AC3	An inactive account sees "Your account is waiting for manager approval".

Status: Not done

US-03 Find a product (Staff)
As staff, I want to search and filter products so that I can check stock while a customer waits.

AC	Acceptance criterion
AC1	The list shows 10 products per page with SKU, name, category, quantity and unit, and page controls.
AC2	Typing "coca" shows only products whose name or SKU contains "coca", ignoring upper and lower case.
AC3	Filters (category, supplier, low stock only) combine with the search; an empty result shows "No products match your search".
AC4	Products at or below their reorder level show a low-stock badge.

Status: Not done

US-04 Record stock in (Staff)

As staff, I want to record a delivery so that the stock count goes up.

AC	Acceptance criterion
AC1	A whole quantity from 1 to 10,000 with reason PURCHASE or CUSTOMER_RETURN is saved, and the product's quantity increases by that amount without a page reload.
AC2	A quantity of 0, a negative number or a decimal shows "Quantity must be a whole number from 1 to 10,000".
AC3	An archived product cannot receive stock; the message "This product is archived" is shown.

Status: Not done

US-05 Record stock out (Staff)

As staff, I want to record items sold or thrown away so that the stock count stays correct.

AC	Acceptance criterion
AC1	A valid stock out with a reason (SALE, DAMAGED, EXPIRED or RETURN_TO_SUPPLIER) decreases the quantity by that amount.
AC2	Taking out more than is in stock is refused with "Not enough stock: only N left", and the quantity does not change.
AC3	If the quantity drops to the reorder level or below, the product appears in the low-stock list at once.

Status: Not done

US-06 See my movements (Staff)

As staff, I want to see the movements I recorded so that I can check my work at the end of a shift.

AC	Acceptance criterion
AC1	My movements page lists only my movements, newest first, filterable by type and date.
AC2	Opening another staff member's movement by URL (/movements/:id) shows the not-found page (the API returns 404).
AC3	A movement has no edit or delete button; mistakes are fixed by a new movement.

Status: Not done

US-07 Manage products (Manager)
As a manager, I want to add, edit and archive products so that the catalogue matches what the shop sells.

AC	Acceptance criterion
AC1	Creating a product with a unique, well-formed SKU adds it to the list with quantity 0.
AC2	A duplicate SKU shows "SKU already exists"; a wrong format shows "SKU must look like DRK-0001".
AC3	A sale price lower than the cost price shows "Sale price cannot be lower than cost price".
AC4	A product with movements cannot be deleted; the message suggests archiving it instead.
AC5	A staff user cannot create, edit or delete products (the API returns 403).

Status: Not done

US-08 Adjust stock after a count (Manager)
As a manager, I want to correct a quantity after counting the shelf so that the system matches reality.

AC	Acceptance criterion
AC1	Entering the counted quantity creates an adjustment with the difference (for example counted 18, system 20, change −2) and a required note.
AC2	A count equal to the current quantity is refused with "No difference to adjust".
AC3	A staff user cannot make an adjustment (the API returns 403).

Status: Not done

US-09 Low-stock alerts and monthly report (Manager)
As a manager, I want to see what is running out and how stock moved this month so that I can reorder in time.

AC	Acceptance criterion
AC1	The alerts page lists products at or below their reorder level, with quantity, reorder level and supplier, most short first.
AC2	The menu shows a badge with the number of low-stock products.
AC3	For a chosen month, the report shows stock in and out per category, the top 5 products sold and the total stock value; a month with no movements shows zeros.

Status: Not done

US-10 Manage staff, categories and suppliers (Manager)
As a manager, I want to control who can use the system and how products are grouped.

AC	Acceptance criterion
AC1	The manager can activate or deactivate a staff account; a deactivated user cannot log in.
AC2	The manager can create, edit and delete categories and suppliers; deleting one that still has products is refused with "Move or delete its products first".
AC3	A staff user cannot change staff accounts, categories or suppliers (the API returns 403).

Status: Not done

9.3 Guard table. Y = allowed; Own = only records the user created (others return 404); — = 403. Without a token, everything except register and log in returns 401.

Action	Staff	Manager
Register, log in	Y	Y
View, search and filter products	Y (active only)	Y
Create, update, archive, delete products	—	Y
Record stock in and stock out	Y	Y
Record an adjustment	—	Y
View movements	Own	Y
View categories and suppliers	Y	Y
Manage categories, suppliers and staff accounts	—	Y
View low-stock alerts and monthly report	—	Y

9.4 Advanced feature limits. Search, filter and pagination on the product list (US-03) is the advanced feature for Check 6. Its limits, which will be tested:

Limit	Expected behaviour
Page size	10 by default; limit above 50 is capped at 50
Page number	page below 1 is refused with 400
Search	Case-insensitive, matches name or SKU; an empty search shows all products
No result	Shows "No products match your search" instead of an empty table